# ANALYTIC SEWING FOR CHIRAL ALGEBRAS: HILBERT–SCHMIDT CRITERIA, BERGMAN REDUCTION, AND THE HEISENBERG FREDHOLM DETERMINANT

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Abstract. We develop an analytic completion of the chiral bar construction that converts formal sewing laws into convergent partition functions. The central object is the sewing envelope $\mathcal{A}^{\mathrm{sew}}$, the universal locally convex completion of an algebraic chiral core for which every bordism amplitude extends continuously. We prove three results. First, under a Hilbert–Schmidt condition on the sector multiplication, every closed amplitude of $\mathcal{A}$ is trace class on a weighted completion $H_q$, for every $0 < q < 1$. Second, for the Heisenberg algebra the sewing envelope on the disk is $\mathrm{Sym}A^2(D)$, the pair-of-pants amplitude is the second quantization of a one-particle Bergman restriction, and every genus-$g$ partition function is a Fredholm determinant on the reduced Bergman space. Third, any positive-energy chiral algebra with subexponential sector growth and polynomial OPE coefficient growth satisfies the HS-sewing condition. The criterion verifies for the Heisenberg, affine Kac–Moody at integer level, Virasoro, $\mathcal{W}_N$, and lattice vertex algebras.

## 1. Introduction

1.1. **Sewing and the bootstrap.** The sewing axiom is the defining structural law of two-dimensional conformal field theory. A Riemann surface with parametrized boundary circles carries an amplitude valued in the tensor power of a state space; gluing two surfaces along matched boundary circles composes the amplitudes; and the conformal bootstrap decomposes higher-genus amplitudes into sphere amplitudes by cutting along every possible configuration of circles. The classical treatment of Segal [5] takes this as the basic geometric axiom. Zhu's modular invariance theorem [6] proves that, for a rational vertex operator algebra, the genus-1 characters span a representation of $SL_2(\mathbb{Z})$; Huang extended the sewing of sphere amplitudes to genus one and then by induction to all genera [2], under a list of analytic hypotheses (convergence of the relevant products of matrix elements; absolute convergence of sector sums).

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For a general chiral algebra the sewing picture is simple to state but analytically delicate. One forms the bar complex $\bar{B}(\mathcal{A})$ of residue-contracted point configurations; genus-$g$ amplitudes are integrals of bar elements over boundary strata of $\overline{\mathcal{M}}_{g,n}$; the gluing of two surfaces is a pair-of-pants multiplication followed by trace. Each of these operations is well-defined at the algebraic level, but asking for a convergent *number* rather than a formal power series requires completing the state space in a topology that both contains the image of every amplitude and supports trace-class estimates. This is the analytic sewing problem.

1.2. **The Hilbert–Schmidt criterion.** We formulate a single analytic hypothesis that subsumes the bootstrap hypotheses of Huang and that can be verified by inspection of OPE data. Let $H = \widehat{\bigoplus}_{n\geq 0} H_n$ be a graded pre-Hilbert core and let $m^c_{a,b}\colon H_a \otimes H_b \to H_c$ denote the sector components of the pair-of-pants product. Say $\mathcal{A}$ satisfies *HS-sewing* if

$$\sum_{a,b,c\geq 0} q^{a+b+c}\, \|m^c_{a,b}\|^2_{\mathrm{HS}} \;<\; \infty \tag{1.1}$$

for some $0 < q < 1$. Our first result (Proposition 3.2) is that this single estimate delivers the full sewing package: every closed amplitude is trace-class on the weighted completion $H_q := \widehat{\bigoplus}_n q^n H_n$, so the genus-$g$ partition functions and correlators are absolutely convergent numbers, not merely formal series.

Our second result verifies the criterion for the Heisenberg algebra in the sharpest possible form. Moriwaki [4] identifies the Heisenberg sewing envelope with the symmetric algebra $\mathrm{Sym}A^2(D)$ of the Bergman space of the disk; we prove that the pair-of-pants amplitude is the second quantization $\Gamma(R)$ of a one-particle Bergman restriction operator $R$, and that every closed amplitude is therefore a Fredholm determinant on a reduced Bergman space. The genus-$g$ partition function is

$$Z_g(\mathcal{H}_k;\,\Omega) \;=\; (\det \operatorname{Im}\Omega)^{-k/2} \cdot (\det{}'_{\zeta}\Delta_{\Sigma_g})^{-k/2}, \tag{1.2}$$

in agreement with the Belavin–Knizhnik [1] string measure.

Our third result turns (1.1) into a general criterion. Any positive-energy chiral algebra with subexponential sector growth $\log\dim H_n = o(n)$ and polynomial OPE coefficient growth $|C^{c,k}_{a,i;b,j}| \leq K(a+b+c+1)^N$ satisfies HS-sewing for every $0 < q < 1$. These are both conditions that can be checked from character formulas and OPE tables. We verify the criterion on the standard landscape: the Heisenberg $\mathcal{H}_k$, affine Kac–Moody $V_k(\mathfrak{g})$ at integer level, Virasoro $\mathrm{Vir}_c$, the $\mathcal{W}_N$ algebras, and lattice vertex algebras $V_\Lambda$.

1.3. **Relation to the monograph and to prior work.** This note is an excerpt from the higher-genus completion chapter of the monograph [3]; we reproduce the three main results in a form suitable for independent citation and verification. The sewing envelope framework is developed in Section 3 of that chapter, the Hilbert–Schmidt criterion in Section 4, and

the Heisenberg Bergman identification in the Heisenberg worked example of the standard-landscape part of the monograph.

Zhu's original sewing for rational VOAs [6] uses modular invariance as its analytic input; Huang's hypotheses for irrational sewing [2] are formulated as convergence of specific formal series. The HS criterion is stronger in that it provides a single norm estimate that implies all such convergences simultaneously. The Bergman identification of the Heisenberg sewing envelope was established by Moriwaki [4] through factorization homology valued in IndHilb; we reinterpret it here as a one-particle reduction of the pair-of-pants operator and deduce the Fredholm expression (1.2).

1.4. **Outline.** Section 2 constructs the sewing envelope as a universal locally convex completion and states its universal property. Section 3 defines the Hilbert–Schmidt sewing condition and proves the trace-class theorem. Section 4 treats the Heisenberg algebra: statement of the sewing theorem, one-particle reduction, and the Fredholm determinant formula. Section 5 proves the general criterion from polynomial OPE growth and subexponential sector growth. Section 6 verifies the criterion on the standard landscape. Section 7 collects open problems.

## 2. The sewing envelope

2.1. **Algebraic cores and bordism amplitudes.** Throughout this paper, an *algebraic chiral core* is a $\mathbb{Z}_{\geq 0}$-graded vector space $\mathcal{A}_{\mathrm{alg}} = \bigoplus_{n\geq 0} H_n$ equipped with the operator product expansion of a vertex operator algebra of positive energy, with $H_0 = \mathbb{C}\cdot\mathbf{1}$ and $\dim H_n < \infty$. The sector components $H_n$ are the conformal-weight eigenspaces; the pair-of-pants product assembles them into the state space. The augmentation $\varepsilon\colon \mathcal{A}_{\mathrm{alg}} \to \mathbb{C}$ is the projection to $H_0$. The augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the input to the bar construction $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

A finite conformally flat bordism is a compact Riemann surface $\Sigma$ with parametrized in/out boundary collars $(D_1^{\mathrm{in}}, \dots, D_m^{\mathrm{in}})$ and $(D_1^{\mathrm{out}}, \dots, D_\ell^{\mathrm{out}})$. For each such $\Sigma$ the algebraic data determines an *algebraic amplitude*

$$A_\Sigma^{\mathrm{alg}}\colon \mathcal{A}_{\mathrm{alg}}^{\otimes m} \longrightarrow \mathcal{A}_{\mathrm{alg}}^{\otimes \ell},$$

a priori valued in formal series of conformal-weight components. Matrix coefficients of $A_\Sigma^{\mathrm{alg}}$ are bilinear pairings

$$\langle \eta,\, A_\Sigma^{\mathrm{alg}}(\xi_1 \otimes \cdots \otimes \xi_m)\rangle \in \mathbb{C}[[\text{moduli of } \Sigma]],$$

where $\eta$ ranges over the algebraic dual of the output sectors. The question is whether these matrix coefficients converge to actual complex numbers.

2.2. **The sewing envelope.**

**Definition 2.1** (Sewing envelope)**.** For every finite conformally flat bordism $\Sigma$ and every choice of boundary vectors $\xi_i \in \mathcal{A}_{\mathrm{alg}}$, $\eta \in \mathcal{A}_{\mathrm{alg}}^\vee$, define the

seminorm

$$p_{\Sigma,\xi,\eta}(a) \;=\; \big|\langle \eta,\, A^{\mathrm{alg}}_{\Sigma}(\xi_1 \otimes \cdots \otimes a \otimes \cdots \otimes \xi_m)\rangle\big|. \tag{2.1}$$

The *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$ is the Hausdorff completion of $\mathcal{A}_{\mathrm{alg}}$ for the locally convex topology generated by all seminorms $p_{\Sigma,\xi,\eta}$.

The seminorms (2.1) demand that every matrix coefficient of every algebraic amplitude extends to a continuous function of the state. The sewing envelope is the smallest completion that remembers every local-to-global operation at once. It is tautologically the correct target for any analytic bootstrap.

**Proposition 2.2** (Universal property)**.** *Let $E$ be any complete locally convex realization of $\mathcal{A}_{\mathrm{alg}}$ for which every algebraic amplitude $A^{\mathrm{alg}}_{\Sigma}$ extends continuously. Then the identity on $\mathcal{A}_{\mathrm{alg}}$ extends uniquely to a continuous map $\mathcal{A}^{\mathrm{sew}} \to E$.*

*Proof.* By hypothesis, each seminorm $p_{\Sigma,\xi,\eta}$ is continuous on $E$. Hence the identity $\mathcal{A}_{\mathrm{alg}} \to E$ is continuous for the sewing topology. By completeness of $E$, it extends uniquely from the Hausdorff completion $\mathcal{A}^{\mathrm{sew}}$. □

*Remark* 2.3 (Algebraic versus analytic completions)*.* The sewing envelope need not coincide with naive completions such as the $q$-adic completion $\widehat{\bigoplus}_n q^n H_n$, the Hilbert-space completion with respect to a fixed invariant form, or Zhu's $C_2$-quotient. All of these are quotients or subspaces of $\mathcal{A}^{\mathrm{sew}}$ under the universal property of Proposition 2.2, but none is obviously large enough to carry every bordism amplitude as a continuous map. The HS-sewing criterion of the next section is the device by which one identifies a concrete Hilbert completion $H_q \subset \mathcal{A}^{\mathrm{sew}}$ and verifies that amplitudes extend continuously to it.

## 3. The Hilbert–Schmidt criterion

The universal sewing envelope of Definition 2.1 is an abstract object. To compute with it one needs a concrete Hilbert completion on which the sewing operators are not merely defined but small enough in norm to compose. The Hilbert–Schmidt criterion is such a device.

**Definition 3.1** (HS-sewing)**.** Let $\mathcal{A}_{\mathrm{alg}} = \bigoplus_n H_n$ be an algebraic chiral core with pre-Hilbert structure on each sector (for example, the invariant form restricted to $H_n$). Write the pair-of-pants sector multiplications as

$$m^c_{a,b}\colon H_a \otimes H_b \longrightarrow H_c.$$

Say $\mathcal{A}$ satisfies *HS-sewing* at parameter $q \in (0,1)$ if

$$\sum_{a,b,c\geq 0} q^{a+b+c}\, \|m^c_{a,b}\|^2_{\mathrm{HS}} \;<\; \infty. \tag{3.1}$$

The *weighted completion* is $H_q := \widehat{\bigoplus}_{n\geq 0} q^n H_n$, with inner product $\langle v, w\rangle_q = q^{2n}\langle v, w\rangle_{H_n}$ for $v, w \in H_n$.

The scaling $q^{2n}$ is the unique convention under which the weighted pair-of-pants $m_q$ is Hilbert–Schmidt precisely when (3.1) is satisfied.

**Proposition 3.2** (Closed amplitudes are trace-class)**.** *Suppose $\mathcal{A}$ satisfies HS-sewing at parameter $q$. Then*

(i) *the weighted pair-of-pants $m_q\colon H_q \otimes H_q \to H_q$ is Hilbert–Schmidt, with*

$$\|m_q\|_{\mathrm{HS}}^2 \;=\; \sum_{a,b,c} q^{a+b+c}\, \|m_{a,b}^c\|_{\mathrm{HS}}^2;$$

(ii) *every closed amplitude of $\mathcal{A}$, built from any finite composition of pair-of-pants multiplications and traces, is trace class on $H_q$;*

(iii) *the sewing envelope $\mathcal{A}^{\mathrm{sew}}$ admits a continuous linear map $\mathcal{A}^{\mathrm{sew}} \to H_q$ whose image is dense.*

*Proof.* Claim (i) is direct calculation. Identifying $H_a \otimes H_b$ with the weight-$(a+b)$ subspace of $H_q^{\widehat{\otimes} 2}$, the sector component $m_{a,b}^c\colon H_a \otimes H_b \to H_c$ contributes $q^a \cdot q^b \cdot q^{-c} \cdot q^{2c} = q^{a+b+c}$ to the Hilbert–Schmidt norm of $m_q$; the $q^{-c}$ factor comes from the output side and the $q^{2c}$ factor from the normalization of the inner product on $H_c$. Summing the sector contributions gives $\|m_q\|_{\mathrm{HS}}^2 = \sum q^{a+b+c} \|m_{a,b}^c\|_{\mathrm{HS}}^2 < \infty$.

Claim (ii) is the multiplicativity of trace-class operators under composition. A genus-$g$ Riemann surface with $n$ punctures admits a pants decomposition into $2g - 2 + n$ pairs of pants; each internal sewing circle corresponds to a Hilbert–Schmidt pair-of-pants composition, and the composition of two Hilbert–Schmidt operators is trace class. Every closed amplitude (an amplitude with no free external legs) therefore evaluates to the trace of a finite composition of trace-class operators.

Claim (iii) is the observation that the seminorms $p_{\Sigma,\xi,\eta}$ of Definition 2.1 are all dominated by the $H_q$ inner product, because each matrix coefficient factorises through the trace-class operator of claim (ii); continuity of the identity $\mathcal{A}_{\mathrm{alg}} \to H_q$ for the sewing topology follows. □

**Corollary 3.3** (Positivity of the sewing Gram matrix)**.** *Under HS-sewing, the connected sewing amplitudes $a_{\mathcal{A}}(N) \in \mathbb{R}_{\geq 0}$ define a positive-semidefinite inner product on generating Dirichlet polynomials. In particular, every closed amplitude lies in the cone of positive-semidefinite finite sums of pair-of-pants amplitudes.*

*Proof.* Trace-class operators form a two-sided ideal with a well-defined trace, which is positive when applied to a positive operator; the sewing amplitude is a composition of positive pair-of-pants operators by the construction of claim (ii) above, and positivity passes through. □

## 4. The Heisenberg sewing theorem

The Heisenberg vertex algebra $\mathcal{H}_k$ is generated by a single current $J(z)$ with operator product expansion

$$J(z)\,J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

Its sectors $H_n$ are spanned by monomials $a_{-n_1-1}\cdots a_{-n_j-1}\mathbf{1}$ with $\sum(n_i+1) = n$; the sector generating function is $\sum_{n\geq 0}\dim H_n\, q^n \;=\; \prod_{n\geq 1}(1-q^n)^{-1}$, equivalently $q^{1/24}/\eta(q)$. The Heisenberg algebra is the first case in which the analytic sewing envelope admits a complete geometric description.

### 4.1. The Bergman identification.

Let $A^2(D)$ be the Bergman space of square-integrable holomorphic functions on the unit disk $D \subset \mathbb{C}$, with inner product $\langle f, g\rangle = \int_D f\overline{g}\,dA$ and orthonormal basis $e_n(z) = \sqrt{(n+1)/\pi}\,z^n$ for $n \geq 0$. The symmetric algebra $\mathrm{Sym}A^2(D)$ is the Hilbert space of symmetric tensors; physically it is the Fock space built on the one-particle Bergman space.

**Definition 4.1** (Mode–Bergman map)**.** The *mode–Bergman map* $\Theta\colon \mathcal{H}_k^{\mathrm{alg}} \longrightarrow \mathrm{Sym}A^2(D)$ is the dense linear map defined on the algebraic Fock core by

$$\Theta(a_{-n-1}\mathbf{1}) \;=\; \sqrt{n+1}\,z^n, \qquad \Theta(a_{-n_1-1}\cdots a_{-n_j-1}\mathbf{1}) \;=\; \mathrm{Sym}\big(\Theta(a_{-n_1-1}\mathbf{1})\otimes\cdots\otimes\Theta(a_{-n_j-1}\mathbf{1})\big).$$

**Theorem 4.2** (Heisenberg sewing)**.** *Let $\mathcal{H}_k$ be the Heisenberg vertex algebra at level $k$ and let $\Theta$ be the mode–Bergman map of Definition 4.1. Then:*

(i) *The sewing envelope $\mathcal{H}_k^{\mathrm{sew}}$ of the algebraic Heisenberg VOA is identified via $\Theta$ with $\mathrm{Sym}A^2(D)$.*
(ii) *The completed bar differential of $\mathcal{H}_k$ on Bergman tensors is the closure of the Gaussian collision operator.*
(iii) *The pair-of-pants sewing amplitude is the second quantization $\Gamma(R)$ of a one-particle Bergman restriction operator $R\colon A^2(D_{\mathrm{out}}) \to A^2(D_{\mathrm{in},1})\otimes A^2(D_{\mathrm{in},2})$.*
(iv) *Every closed genus-$g$ amplitude is a Fredholm determinant on the reduced one-particle Bergman space $A^2_0(\partial D)$ (constant mode removed).*

*Proof.* Claim (i) is the Bergman identification of Moriwaki [4], who constructs a conformally flat 2-disk algebra structure on $\mathrm{Sym}A^2(D)$ isomorphic to the Heisenberg sewing completion by factorization homology valued in IndHilb. The universal property of Proposition 2.2 identifies his completion with $\mathcal{H}_k^{\mathrm{sew}}$.

Claim (ii) is the analytic-algebraic comparison theorem: the algebraic bar differential, written in the monomial basis of the Heisenberg Fock space, is

$$d_{\mathrm{bar}}(a_{-n-1}\mathbf{1}\otimes a_{-m-1}\mathbf{1}) \;=\; k\delta_{n+m+2,\,0}\,\mathbf{1},$$

(i.e. the simple-pole Wick contraction of the generating current). Under $\Theta$, this contraction is the Gaussian collision operator $G(z,w) = k/(1-z\overline{w})^2$ on $A^2(D)\otimes A^2(D)$, which is the reproducing kernel of the Bergman space; the

bar differential closes to the densely defined Gaussian kernel operator on Bergman tensors.

Claims (iii) and (iv) follow from Theorem 4.3 below. □

4.2. **One-particle sewing.** The Heisenberg amplitude factorises completely through a one-particle operator; this is the analytic version of Wick's theorem.

**Theorem 4.3** (Heisenberg one-particle sewing)**.** *The Heisenberg pair-of-pants amplitude is the second quantisation $\Gamma(R)$ of the one-particle Bergman restriction $R\colon A^2(D_{\mathrm{out}}) \to A^2(D_{\mathrm{in},1}) \otimes A^2(D_{\mathrm{in},2})$ whose matrix elements satisfy*

$$|R_{n,m}| \;\le\; Cq^{(n+m)/2},$$

*where $q = e^{-t}$ and $t$ is the conformal length of the collar. The sewing operator $T_q$ on the reduced Bergman space $A^2_0(\partial D)$ has eigenvalues $q^n$ for $n \ge 1$ and is trace class with*

$$\|T_q\|_1 \;=\; \frac{q}{1-q}.$$

*The genus-$g$ partition function of $\mathcal{H}_k$ is the Fredholm determinant*

$$Z_g(\mathcal{H}_k;\,\Omega) \;=\; (\det \operatorname{Im}\Omega)^{-k/2} \cdot (\det{}'_\zeta \Delta_{\Sigma_g})^{-k/2}. \tag{4.1}$$

*At genus 1, $Z_1 = (\operatorname{Im}\tau)^{-k/2}\,|\eta(\tau)|^{-2k}$.*

*Proof.* Wick's theorem: every Heisenberg amplitude factors into pairwise contractions of the generating current. Each contraction across a sewing circle corresponds to a one-particle pairing, so the full Fock-space amplitude equals the second quantization $\Gamma(R)$ of the one-particle pairing $R$.

The one-particle matrix element is

$$R_{n,m} \;=\; \frac{k}{2\pi i}\oint_{|z|=1}\oint_{|w|=1} z^{-n-1}\,w^{-m-1}\,G_{\mathrm{pants}}(z,w)\,dz\,dw,$$

where $G_{\mathrm{pants}}$ is the Bergman kernel of the pair of pants. The collar of length $t = -\log q$ and the Bergman reproducing property give the estimate $|G_{\mathrm{pants}}(z,w)| \le C(1-q|zw|)^{-2}$, and extraction of the $(n,m)$ Fourier coefficient yields $|R_{n,m}| \le Cq^{(n+m)/2}$.

Consequently $R$ is Hilbert–Schmidt, so the second-quantised operator $\Gamma(R)$ is trace-class, and $\|\Gamma(R)\|_1 = \det(1+|R|)^k < \infty$ for all $k \ge 0$.

The Polyakov formula for the Heisenberg partition function identifies $\det(1-K_g)^{-k} = (\det \operatorname{Im}\Omega)^{-k/2}(\det{}'_\zeta \Delta_{\Sigma_g})^{-k/2}$, where $K_g$ is the genus-$g$ sewing kernel assembled by composing $2g-2$ pair-of-pants restriction operators along the edges of a trivalent graph $\Gamma$ of first Betti number $g$; this is the Belavin–Knizhnik [1] holomorphic factorisation of the bosonic string measure.

At genus 1, $K_1 = T_q$ and the Fredholm determinant is $\det(1-T_q) = \prod_{n\ge1}(1-q^n) = q^{-1/24}\eta(q)$, so $Z_1 = (\operatorname{Im}\tau)^{-k/2}\cdot|\eta(\tau)|^{-2k}$ after combining holomorphic and antiholomorphic sectors. □

*Remark* 4.4. The Heisenberg algebra is the unique member of the standard landscape in which every step of the sewing programme is explicit: the sewing envelope is a classical Bergman space; the pair-of-pants amplitude is a Fock-space second quantization; and every partition function is a Fredholm determinant. All non-abelian chiral algebras known to satisfy HS-sewing require the abstract approach of Section 5.

## 5. The general HS-sewing theorem

For algebras beyond the Heisenberg, the sewing envelope does not admit a classical description, but the Hilbert–Schmidt criterion can still be verified from two OPE-level estimates: polynomial growth of the structure constants and subexponential growth of the sector dimensions. Both hypotheses can be read off from character formulas and OPE tables.

**Theorem 5.1** (General HS-sewing criterion)**.** *Let $\mathcal{A}$ be a positive-energy chiral algebra with a choice of pre-Hilbert structure on each sector $H_n$. Suppose*

(i) ***(subexponential sector growth)*** $\log \dim H_n = o(n)$*, equivalently* $\dim H_n \le Ce^{\alpha\sqrt{n}}$ *for some* $C, \alpha > 0$*;*

(ii) ***(polynomial OPE coefficient growth)*** *there exist constants* $K > 0$ *and* $N \ge 0$ *such that the matrix elements of* $m^c_{a,b}$ *in orthonormal bases satisfy*

$$|C^{c,k}_{a,i;\,b,j}| \;\le\; K(a+b+c+1)^N$$

*for all weights* $a, b, c$ *and all basis indices* $i, j, k$*.*

*Then $\mathcal{A}$ satisfies HS-sewing at every parameter* $0 < q < 1$*.*

*Proof.* From hypothesis (ii), the Hilbert–Schmidt norm of the sector multiplication is controlled by

$$\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\le\; \dim H_a \cdot \dim H_b \cdot \dim H_c \cdot K^2(a+b+c+1)^{2N}. \tag{5.1}$$

Apply the power-mean inequality $(x+y+z)^p \le 3^{p-1}(x^p+y^p+z^p)$ with $p = 2N$ and $x = a+1$, $y = b+1$, $z = c+1$:

$$(a{+}b{+}c{+}1)^{2N} \;\le\; (a{+}1{+}b{+}1{+}c{+}1)^{2N} \;\le\; 3^{2N-1}\big((a{+}1)^{2N}{+}(b{+}1)^{2N}{+}(c{+}1)^{2N}\big).$$

Substituting into (5.1) and using the symmetry of the three terms, the HS-sewing sum is bounded by

$$\sum_{a,b,c\ge 0} q^{a+b+c}\,\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\le\; 3^{2N}K^2\Big(\sum_{n\ge 0} q^n \dim H_n\,(n{+}1)^{2N}\Big)\Big(\sum_{n\ge 0} q^n \dim H_n\Big)^2.$$

By hypothesis (i), $\dim H_n \le Ce^{\alpha\sqrt{n}}$, while $q^n = e^{-n\log(1/q)}$ decays exponentially. Each factor $q^n \dim H_n\,(n{+}1)^{2N}$ is bounded by $C(n{+}1)^{2N}\exp(\alpha\sqrt{n} - n\log(1/q))$, which decays superpolynomially, so each series converges absolutely. □

**Corollary 5.2** (Fredholm determinant formula)**.** *Suppose $\mathcal{A}$ satisfies the hypotheses of Theorem 5.1. Then for every stable pants decomposition of a Riemann surface $\Sigma_{g,n}$ of type $(g,n)$ the corresponding closed amplitude admits an absolutely convergent Fredholm expansion*

$$\langle A_{\Sigma_{g,n}} \rangle \;=\; \sum_{\text{trees}\,\tau} \mathrm{Tr}(m_q^\tau),$$

*with each summand a trace of a trace-class operator on $H_q$.*

*Proof.* Immediate from Proposition 3.2 applied to the pants decomposition and the fact that the trace of a trace-class operator is a norm-convergent sum over any orthonormal basis. □

*Remark* 5.3 (Completed algebras). The standard families below are all finitely strongly generated in the sense that a finite set of fields produces the entire algebra under normal-ordered product and derivation. For completed algebras (such as $\mathcal{W}_{1+\infty}$ or completed Yangians), the polynomial OPE bound of hypothesis (ii) must be verified in the completed (inverse limit) topology, where the structure constants are themselves formal power series. This verification is the content of Theorem MC4 of the monograph [3].

## 6. Verification on the standard landscape

We verify the hypotheses of Theorem 5.1 for the standard families. The data required is a character formula (to control $\dim H_n$) and a polynomial OPE bound (to determine $N$).

**Example 6.1** (Heisenberg)**.** The Heisenberg $\mathcal{H}_k$ has $\dim H_n = p(n)$, the partition number. The OPE $J(z)J(w) \sim k/(z-w)^2$ has structure constants bounded by $k$, independent of weight; taking $N = 1$ is safe (the derivation $\partial$ increases weight by 1, so the normal-ordered products of derivatives are at most linear in the degree). Hence $\mathcal{H}_k$ satisfies HS-sewing at every $0 < q < 1$. The sector sum $\sum_n q^n p(n) = \prod_{n\geq 1}(1-q^n)^{-1} = q^{1/24}/\eta(q)$ is the Heisenberg partition generating function, consistent with (4.1).

**Example 6.2** (Affine Kac–Moody at integer level)**.** Let $V_k(\mathfrak{g})$ be the vacuum module of the affine vertex algebra at integer level $k \in \mathbb{Z}_{>0}$, with $\mathfrak{g}$ a simple Lie algebra. The sectors satisfy $\dim H_n \leq Ce^{\alpha\sqrt{n}}$ from the Kac–Weyl character formula. The OPE of two currents $J^a(z)J^b(w)$ has a double pole (central term) and a simple pole (Lie bracket); hence $|C^{c,k}_{a,i;\,b,j}| \leq K(n+1)$ with $N = 1$, because the structure constants depend linearly on the weight via the insertion of derivatives. HS-sewing at every $0 < q < 1$ follows. The sector sum $\sum_n q^n \dim H_n$ is the Kac–Moody character $\chi_{V_k}(q)$.

**Example 6.3** (Virasoro)**.** The Virasoro algebra $\mathrm{Vir}_c$ has sectors with $\dim H_n = p(n)$. The OPE $T(z)T(w) \sim c/2(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$ is quartic; repeated normal products of $T$ give $|C| \leq K(n+1)^2$ with $N = 2$. The general Theorem 5.1 applies for every $0 < q < 1$; the

sector sum $\sum_n q^n p(n) = q^{(1-c)/24}/\eta(q)$ is the Virasoro vacuum character. Note: the criterion applies at every central charge $c$; the specialization $c < 1$ discussed in Zhu [6] corresponds to minimal-model rationality, which the HS criterion does not require.

**Example 6.4** ($\mathcal{W}_N$-algebras)**.** The principal $\mathcal{W}_N$-algebra is generated by $N-1$ primary fields of conformal weights $2, 3, \ldots, N$. Sector dimensions satisfy $\dim H_n \leq Ce^{\alpha\sqrt{n}}$ from the $\mathcal{W}_N$ character formula. OPE coefficient growth is $|C| \leq K(n+1)^{2N}$; applying Theorem 5.1 with this $N$ gives HS-sewing at every $0 < q < 1$, and sector summation produces the $\mathcal{W}_N$ character $\chi_{\mathcal{W}_N}(q)$.

**Example 6.5** (Lattice vertex algebras)**.** Let $\Lambda$ be a positive-definite even lattice of rank $r$ with dual $\Lambda^*$. The lattice vertex algebra $V_\Lambda$ has sectors $\dim H_n = |\Lambda^*/\Lambda| \cdot p_r(n)$, where $p_r$ is the $r$-coloured partition number. The OPE is a finite sum of vertex operators labelled by lattice vectors, so $|C| \leq K \cdot r$ independent of weight, giving $N = r$. HS-sewing at every $0 < q < 1$ follows, and sector summation produces the lattice character $\Theta_\Lambda(q)/\eta(q)^r$.

**Example 6.6** (Summary)**.** The following table summarizes the verification:

| $\mathcal{A}$ | $\dim H_n$ | $\lvert C\rvert$ | $N$ | $\sum_n q^n \dim H_n$ |
|---|---|---|---|---|
| $\mathcal{H}_k$ | $p(n)$ | $\leq k$ | 1 | $q^{1/24}/\eta(q)$ |
| $V_k(\mathfrak{g})$ | $\leq Ce^{\alpha\sqrt{n}}$ | $\leq K(n+1)$ | 1 | $\chi_{V_k}(q)$ |
| $\mathrm{Vir}_c$ | $p(n)$ | $\leq K(n+1)^2$ | 2 | $q^{(1-c)/24}/\eta(q)$ |
| $\mathcal{W}_N$ | $\leq Ce^{\alpha\sqrt{n}}$ | $\leq K(n+1)^{2N}$ | $2N$ | $\chi_{\mathcal{W}_N}(q)$ |
| $V_\Lambda$ | $\lvert\Lambda^*/\Lambda\rvert \cdot p_r(n)$ | $\leq K \cdot r$ | $r$ | $\Theta_\Lambda(q)/\eta(q)^r$ |

Every entry satisfies the hypotheses of Theorem 5.1, so every standard family admits a Hilbert–Schmidt sewing envelope with convergent higher-genus partition functions.

## 7. Conclusion and open problems

The results above upgrade the algebraic MC element of the chiral bar construction to an analytic object that produces genuine partition functions. The Hilbert–Schmidt criterion of Definition 3.1 is verifiable by inspection of OPE data and sector growth; the closed amplitudes of any HS-sewing chiral algebra are trace class on a weighted Hilbert completion $H_q$; for the Heisenberg algebra, the entire sewing programme is explicit in a single Bergman space, and every partition function is a Fredholm determinant.

Three problems stand out.

Completed algebras. The monograph's MC4 completion programme handles inverse-limit algebras such as $\mathcal{W}_{1+\infty}$ and completed Yangians. A quantitative HS criterion in the inverse-limit topology is needed to extend Theorem 5.1 to that setting; see Remark 5.3 and [3] for the algebraic input.

Fredholm determinants beyond Heisenberg. For non-abelian Kac–Moody and Virasoro, the closed amplitudes are trace class but are not yet known to be Fredholm determinants of a single one-particle operator. The Heisenberg identification suggests the existence of an analogous second-quantization structure for every chiral algebra with quadratic OPE; finding the correct one-particle Hilbert space is the natural next step.

Positivity and modular invariance. Corollary 3.3 establishes a positivity cone for HS-sewing amplitudes; it does not directly imply $SL_2(\mathbb{Z})$ covariance of genus-1 characters. Combining the analytic sewing of this paper with Zhu's modular invariance theorem [6] and Huang's higher-genus extension [2] is the natural route to a unified analytic modular programme.

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